



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE-A

Course Code & Title: CS3452 Theory of Computation

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit IV / Pumping Lemma for CFL

Course Outcome: CO 4

Topic Learning Outcome: TLO 13

Activity Chosen: JFLAP-Interactive Visualization Tool

Justification:

- JFLAP is an interactive software tool designed to help students visualize and experiment with concepts in automata theory, formal languages, and computational theory. Using JFLAP supports active learning by allowing students to construct, simulate, and test finite automata, pushdown automata, Turing machines. This hands-on experience deepens their conceptual understanding by transforming abstract theoretical ideas into concrete visual models. By exploring JFLAP independently, students develop problem-solving skills and gain immediate feedback on their designs, which promotes self-directed learning. Additionally, the tool helps clarify complex topics such as automata minimization, language recognition, and the pumping lemma, making these concepts easier to grasp and less intimidating.
- **Time Allotted for the Activity:** 30 minutes

Details of the Implementation:

- **Preparation and Instructions:**
Students of the class were informed in advance to bring their laptops for the session.
- **Installation of JFLAP:**
The faculty guided the students to download and install the JFLAP software on their laptops.
- **Introduction to the Tool:**
Brief instructions were given on the basic features and purpose of JFLAP as an interactive visualization tool for automata theory.
- **Independent Exploration:**
Students were encouraged to explore the tool independently, experimenting with building and simulating different automata models.
- **Group Formation:**
Students were divided into small groups of 7-8 members to facilitate peer learning and discussion.

- **Faculty Interaction and Support:**

The faculty visited each group, answered questions, provided clarifications, and demonstrated key concepts, particularly focusing on the Pumping Lemma.

CO – PO / PSO mapping:

CO	PO1	PO2	PO4	PO9	PO10	PSO1
CO 1	2	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO9	PO10	PSO1
	2	2	1	1	1	1
Justification for correlation	Students used mathematical knowledge to understand and solve problems involving the Pumping Lemma.	Students learned to identify problems that can be solved by analyzing languages using the Pumping Lemma.	Students explored the importance of the Pumping Lemma to check if a language is context or not.	Worked as an individual.	Communicated effectively on complex engineering activities.	Students will be able to identify and formulate the appropriate grammar for the language.

- **Images / Screenshot of the practice:**



Figure 1: Teacher-Student Interaction



Figure 2: Faculty Facilitation Snapshot

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- Students appreciated the hands-on experience, stating that the interactive nature of JFLAP helped them understand pumping lemma concepts more clearly.
- Many students highlighted that visualizing abstract ideas through the tool made learning engaging and easier.
- Instructor noted increased student participation and enthusiasm during the sessions.

❖ **Benefit of the practice:**

- This activity helped the students actively participated and collaborated, fostering teamwork and peer learning.
- The approach fostered self-directed learning, ensuring that students took ownership of their learning process.
- This activity encouraged the students to share their knowledge with others.
- Students learnt at their own pace, revisiting concepts as needed using the JFLAP tool.

❖ **Challenges faced in implementation:**

- Some students faced difficulties installing software due to low internet connectivity.
- Not all students consistently had access to a laptop during the sessions.

References:

1. <https://www.jflap.org/jflaptmp/>
2. <https://jflap.org/jflaptmp/toRun.html>
3. <https://www.youtube.com/watch?reload=9&v=L4KFLmA1X30&pp=ygUGI2pmbGFw>
4. <https://www.rose-hulman.edu/class/csse/resources/JFLAP/index.html>
5. <https://www.yellowfinbi.com/blog/benefits-of-data-visualization-tools>